

AI Defence Lab: Adaptive Experiment

QwenAgents + conditional CTGAN · 3 seeds x 5 rounds · Kaggle GPU

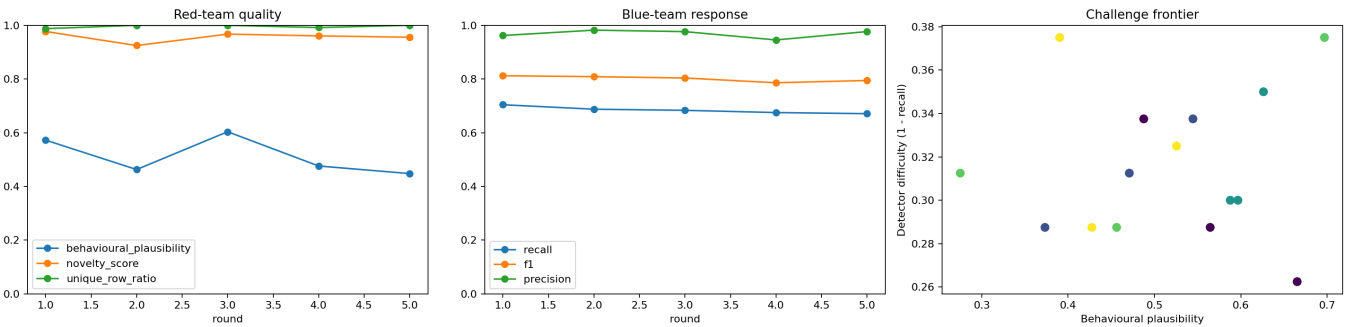
Adaptive method

Round 1 begins with the seeded approved-family plan. After each detector evaluation, Agent 3 records a weakness. Agent 1 uses that weakness and recent memory to recommend the next approved family. The controller validates the recommendation, Agent 2 writes structured constraints, and CTGAN generates strictly family-pure synthetic rows.

Decision trace

Seed	Round	Family	Decision source
20260821	1	account_takeover	seeded_plan
20260821	2	trusted_device	agent_1_adaptive_recommendation
20260821	3	beneficiary_manipulation	agent_1_adaptive_recommendation
20260821	4	low_and_slow	agent_1_adaptive_recommendation
20260821	5	cross_channel_anomaly	agent_1_adaptive_recommendation
20260822	1	account_takeover	seeded_plan
20260822	2	trusted_device	agent_1_adaptive_recommendation
20260822	3	beneficiary_manipulation	agent_1_adaptive_recommendation
20260822	4	low_and_slow	agent_1_adaptive_recommendation
20260822	5	social_engineering	agent_1_adaptive_recommendation
20260823	1	social_engineering	seeded_plan
20260823	2	account_takeover	agent_1_adaptive_recommendation
20260823	3	trusted_device	agent_1_adaptive_recommendation
20260823	4	low_and_slow	agent_1_adaptive_recommendation
20260823	5	beneficiary_manipulation	agent_1_adaptive_recommendation

Red team and blue team behaviour



Metric	Mean	Std	Min	Max
f1	0.801	0.0205	0.7634	0.8321
recall	0.6842	0.0324	0.625	0.7375
precision	0.9685	0.0281	0.8906	1.0
roc_auc	0.9248	0.0185	0.8821	0.9559
novelty_score	0.9569	0.0256	0.8758	1.0
behavioural_plausibility	0.5126	0.1127	0.2751	0.6969

Interpretation

The red team maintained measurable novelty and behavioural plausibility while changing families in response to weaknesses. The blue team was evaluated on unseen attacks; recall and F1 describe detection response, while false-positive control remains part of the evaluation protocol. These are internal synthetic-experiment indicators, not official Mastercard scores or live-payment performance claims.